

**Section A: Pure Mathematics**

| No.  | Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1i)  | $x = 3t^2, \quad y = 6t$<br>$\frac{dx}{dt} = 6t, \quad \frac{dy}{dt} = 6$<br>$\Rightarrow \frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx} = \frac{1}{t}$<br>$\frac{dy}{dx} = 0.4 \Rightarrow t = 2.5$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Those who tried to obtain Cartesian equation of curve $C$ first usually made errors.                                                                                                                                                                                                                                                                                                                                                                                       |
| 1ii) | <p>At <math>P(3p^2, 6p)</math>, <math>6t = 6p \Rightarrow t = p</math></p> <p>Equation of tangent at the point <math>P(3p^2, 6p)</math> are given by</p> $y - 6p = \frac{1}{p}(x - 3p^2)$ <p><math>x = 0, y = 6p - 3p = 3p \Rightarrow</math> coordinates of <math>D</math> is <math>(0, 3p)</math></p> <p>Coordinates of mid-point of <math>PD</math> is <math>\left(\frac{0+3p^2}{2}, \frac{3p+6p}{2}\right)</math></p> <p style="text-align: center;">i.e. <math>\left(\frac{3p^2}{2}, \frac{9p}{2}\right)</math></p> <p>As <math>p</math> varies, <math>x = \frac{3p^2}{2}</math> and <math>y = \frac{9p}{2}</math></p> <p><math>\therefore x = \frac{3}{2}\left(\frac{2y}{9}\right)^2 \Rightarrow 27x = 2y^2</math></p>                                                                                                                                                                                                                                                                                                                      | <p>Locus: curve traced out by a variable point. In this question, the locus we want to find is the curve traced out by the mid-point of <math>PD</math> as <math>p</math> varies</p> <p><u>Similar to 2011/P1/Q3 (iii)</u></p> <p>Still only a very small number of candidates were able to find the Cartesian equation of the locus correctly after obtaining the correct coordinates of the mid-point of <math>PQ</math>.</p>                                            |
| 2    | <p>Let <math>\frac{9x^2+x-13}{(2x-5)(x^2+9)} = \frac{A}{2x-5} + \frac{Bx+C}{x^2+9} \quad \text{---(a)}</math></p> <p><math>\therefore A(x^2+9) + (Bx+C)(2x-5) = 9x^2+x-13</math></p> <p><math>A+2B=9 \quad \text{---(1)}</math></p> <p><math>-5B+2C=1 \quad \text{---(2)}</math></p> <p><math>9A-5C=-13 \quad \text{---(3)}</math></p> <p>Solving the 3 equations using GC: <math>A=3, B=3, C=8</math></p> $\int_0^2 \frac{9x^2+x-13}{(2x-5)(x^2+9)} dx = \int_0^2 \left( \frac{3}{2x-5} + \frac{3x+8}{x^2+9} \right) dx$ $= \int_0^2 \left[ \frac{3}{(2x-5)} + \frac{3x}{x^2+9} + \frac{8}{x^2+9} \right] dx$ $= \left[ \frac{3}{2} \ln 2x-5  + \frac{3}{2} \ln(x^2+9) + \frac{8}{3} \tan^{-1}\left(\frac{x}{3}\right) \right]_0^2$ $= 0 + \frac{3}{2} \ln 13 + \frac{8}{3} \tan^{-1}\left(\frac{2}{3}\right) - \frac{3}{2} \ln 5 - \frac{3}{2} \ln 9 - 0$ $= \frac{3}{2} \ln \frac{13}{45} + \frac{8}{3} \tan^{-1}\left(\frac{2}{3}\right)$ <p><math>\therefore a = \frac{3}{2}, b = \frac{13}{45}, c = \frac{8}{3}, d = \frac{2}{3}</math></p> | <p>Well-done.</p> <p>However there are many places where students can make mistakes.</p> <ol style="list-style-type: none"> <li><math>\frac{Bx+C}{x^2+9}</math> written as <math>\frac{B}{x^2+9}</math> instead.</li> <li><math>\int_0^2 \frac{3x+8}{x^2+9} dx</math> did not split the numerator when integrating.</li> <li><math>\frac{3}{2} \ln \frac{13}{45} + \frac{8}{3} \tan^{-1}\left(\frac{2}{3}\right) + C</math> although it is a definite integral.</li> </ol> |

3ia)

The distance completed in Version 1 forms an A.P. with  $a = 8$  and common difference,  $d = 8$ .

|          |                             |
|----------|-----------------------------|
| Stage 1  | $u_1 = 2 \times 4 = 8$      |
| Stage 2  | $u_2 = 2 \times 8 = 16$     |
| Stage 3  | $u_3 = 2 \times 12 = 24$    |
| $\vdots$ | $\vdots$                    |
| Stage 10 | $u_{10} = 8 + 8(10-1) = 80$ |

Distance completed in the first 10 stages

$$= \frac{10}{2} [2 \times 8 + (10-1) \times 8] = 440 \text{ m}$$

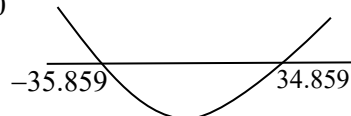
3ib)

Distance completed in the first  $n$  stages

$$= \frac{n}{2} [2 \times 8 + (n-1) \times 8] = 4n^2 + 4n$$

To complete at least 5 km,  $4n^2 + 4n \geq 5000$

$$\therefore n^2 + n - 1250 \geq 0$$



$$\Rightarrow n \geq 34.859 \text{ or } n \leq -35.859 \text{ (rejected as } n > 0)$$

Least number of stages is 35.

3ii)

|           |                                |
|-----------|--------------------------------|
| Stage 1   | $u_1 = 8$                      |
| Stage 2   | $u_2 = 2(8)$                   |
| Stage 3   | $u_3 = 2(2 \times 8) = 2^2(8)$ |
| $\vdots$  | $\vdots$                       |
| Stage $n$ | $u_n = 2^{n-1}(8)$             |

The distance completed in Version 2 forms a G.P. with  $a = 8$  and common difference,  $d = 2$ .

Distance completed in the first  $n$  stages

$$= \frac{8(2^n - 1)}{2 - 1} = 2^{n+3} - 8$$

$$2^{n+3} - 8 = 10000$$

$$n + 3 = \frac{\ln 10008}{\ln 2} \Rightarrow n \approx 10.289$$

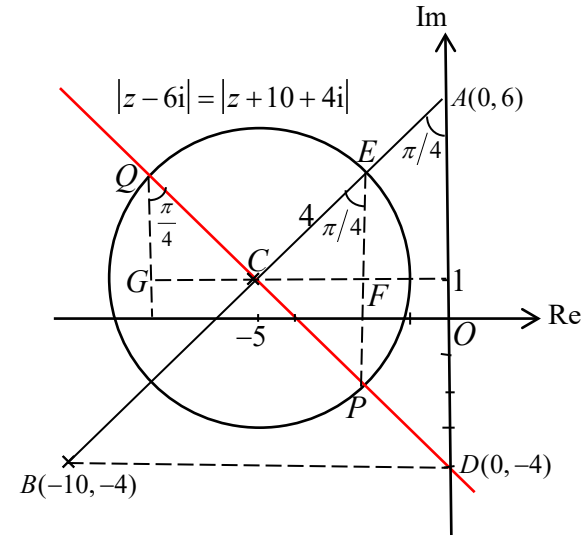
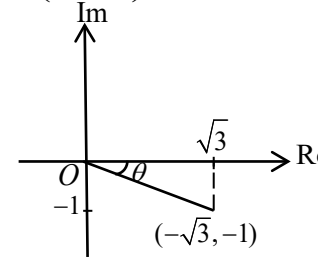
Distance completed in the first 10 stages

$$= 2^{13} - 8 = 8184 \text{ m}$$

To run exactly 10 km, the athlete has to run another  $10000 - 8184 = 1816 \text{ m}$

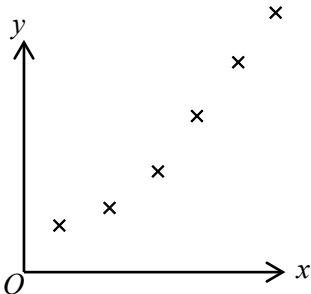
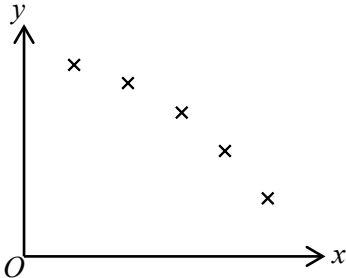
$$\text{Distance to run from } O \text{ to } A_{11} = \frac{2^{10}(8)}{2} = 4096$$

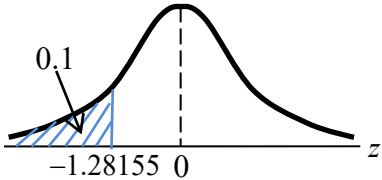
Since  $1816 < 4096$ , hence athlete has to run away from  $O$ , in the direction from  $O$  to  $A$ .

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                              |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| 4ai)  |  <p>Diagram illustrating the locus of <math>z</math> in the complex plane. The locus is a circle centered at <math>C(-5, 1)</math> with radius 4. A red line passes through points <math>A(0, 6)</math> and <math>B(-10, -4)</math>. The circle intersects this line at points <math>P</math> and <math>Q</math>. Dashed lines connect <math>C</math> to the axes and to points <math>E</math> and <math>F</math> on the circle. Angles of <math>\pi/4</math> are marked at <math>A</math> and <math>B</math>.</p> | Not in H2 Math 9758 syllabus |
| 4aii) | $ z - 6i  =  z + 10 + 4i $<br>Locus of $z$ is the perpendicular bisector of the line joining the points $A(0, 6)$ and $B(-10, -4)$ .<br>$CF = 4 \cos \frac{\pi}{4} = 2\sqrt{2} = CG$<br>$EF = FP = 4 \sin \frac{\pi}{4} = 2\sqrt{2} = QG$<br>The complex numbers representing the 2 points $P$ and $Q$ are $(-5 + 2\sqrt{2}) + (1 - 2\sqrt{2})i$ and $(-5 - 2\sqrt{2}) + (1 + 2\sqrt{2})i$ .                                                                                                                                                                                                        | Not in H2 Math 9758 syllabus |
| 4bi)  | $w = (\sqrt{3}) - i = 2e^{i(-\frac{\pi}{6})}$<br>$w^6 = \left(2e^{i(-\frac{\pi}{6})}\right)^6 = 2^6 e^{i(-\pi)} = 64e^{i\pi}$<br> <p>Diagram illustrating the complex number <math>w</math> in the complex plane. The point <math>w</math> is at <math>(-\sqrt{3}, -1)</math>. The angle <math>\theta</math> is measured from the positive real axis to the vector representing <math>w</math>.</p>                                                                                                              |                              |
| 4bii) | $\frac{w^n}{w^*} = \frac{2^n e^{i(-\frac{n\pi}{6})}}{2e^{i(\frac{\pi}{6})}} = 2^{n-1} e^{i(-\frac{n\pi}{6} - \frac{\pi}{6})}$<br>$= 2^{n-1} \left[ \cos\left(-\frac{n\pi}{6} - \frac{\pi}{6}\right) + i \sin\left(-\frac{n\pi}{6} - \frac{\pi}{6}\right) \right]$<br>For $\frac{w^n}{w^*}$ to be a real number, $\sin\left(-\frac{n\pi}{6} - \frac{\pi}{6}\right) = 0$<br>$\therefore -\frac{n\pi}{6} - \frac{\pi}{6} = -\pi, -2\pi, -3\pi, \dots$<br>$\therefore$ the 3 smallest positive values of $n = 5, 11, 17$                                                                                |                              |

**Section B: Statistics**

| No                          | Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Comments                     |                 |                             |                                                                    |                             |                                                                     |                      |                                                               |  |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|-----------------|-----------------------------|--------------------------------------------------------------------|-----------------------------|---------------------------------------------------------------------|----------------------|---------------------------------------------------------------|--|
| 5i)                         | Random interval $\frac{10000}{5\% \times 10000} = 20$ .<br>Each of the 10000 customers in the list compiled by the internet retailer is assigned a number from 1 to 10000. Randomly select a number from 1 to 20, say 6. Then select the first customer numbered 6 and subsequently select every 20 <sup>th</sup> customer (i.e. 26 <sup>th</sup> , 46 <sup>th</sup> , 66 <sup>th</sup> , ...) until a total of 5% of the customers (500) are selected.                                                                                                                                                                                                           | Not in H2 Math 9758 syllabus |                 |                             |                                                                    |                             |                                                                     |                      |                                                               |  |
| 5ii)                        | The advantage is that the selection is spread out over the 10000 regular customers. The disadvantage is that the sample may comprise similar characteristics of customers if there is periodicity in the compiled list. For example, every 20 <sup>th</sup> customer happens to be a male in a certain age group depending on how the list is compiled.                                                                                                                                                                                                                                                                                                           | Not in H2 Math 9758 syllabus |                 |                             |                                                                    |                             |                                                                     |                      |                                                               |  |
| 6i)                         | Number of teams<br>$= {}^3C_1 \times {}^8C_4 \times {}^5C_2 \times {}^6C_4$<br>$= 31500$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                              |                 |                             |                                                                    |                             |                                                                     |                      |                                                               |  |
| 6ii)                        | Suppose the 2 brothers are $A$ (midfielder) and $B$ (attacker). <table border="1"><thead><tr><th>Team composition</th><th>Number of teams</th></tr></thead><tbody><tr><td><math>A</math> is in and <math>B</math> is not in</td><td><math>{}^3C_1 \times {}^8C_4 \times {}^4C_1 \times {}^5C_4</math><br/><math>= 4200</math></td></tr><tr><td><math>A</math> is not in and <math>B</math> is in</td><td><math>{}^3C_1 \times {}^8C_4 \times {}^4C_2 \times {}^5C_3</math><br/><math>= 12600</math></td></tr></tbody></table><br>Required number of teams = 16800                                                                                                 | Team composition             | Number of teams | $A$ is in and $B$ is not in | ${}^3C_1 \times {}^8C_4 \times {}^4C_1 \times {}^5C_4$<br>$= 4200$ | $A$ is not in and $B$ is in | ${}^3C_1 \times {}^8C_4 \times {}^4C_2 \times {}^5C_3$<br>$= 12600$ |                      |                                                               |  |
| Team composition            | Number of teams                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                              |                 |                             |                                                                    |                             |                                                                     |                      |                                                               |  |
| $A$ is in and $B$ is not in | ${}^3C_1 \times {}^8C_4 \times {}^4C_1 \times {}^5C_4$<br>$= 4200$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                              |                 |                             |                                                                    |                             |                                                                     |                      |                                                               |  |
| $A$ is not in and $B$ is in | ${}^3C_1 \times {}^8C_4 \times {}^4C_2 \times {}^5C_3$<br>$= 12600$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                              |                 |                             |                                                                    |                             |                                                                     |                      |                                                               |  |
| 6iii)                       | Both $A$ and $B$ leave the club. Suppose $C$ can play either a midfielder or a defender. <table border="1"><thead><tr><th>Team composition</th><th>Number of teams</th></tr></thead><tbody><tr><td><math>C</math> is not included</td><td><math>{}^3C_1 \times {}^8C_4 \times {}^3C_2 \times {}^5C_4 = 3150</math></td></tr><tr><td><math>C</math> plays defender</td><td><math>{}^3C_1 \times {}^8C_3 \times {}^3C_2 \times {}^5C_4 = 2520</math></td></tr><tr><td><math>C</math> plays midfielder</td><td><math>{}^3C_1 \times {}^8C_4 \times {}^3C_1 \times {}^5C_4 = 3150</math></td></tr></tbody></table><br>Required number of teams = 3150+2520+3150=8820. | Team composition             | Number of teams | $C$ is not included         | ${}^3C_1 \times {}^8C_4 \times {}^3C_2 \times {}^5C_4 = 3150$      | $C$ plays defender          | ${}^3C_1 \times {}^8C_3 \times {}^3C_2 \times {}^5C_4 = 2520$       | $C$ plays midfielder | ${}^3C_1 \times {}^8C_4 \times {}^3C_1 \times {}^5C_4 = 3150$ |  |
| Team composition            | Number of teams                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                              |                 |                             |                                                                    |                             |                                                                     |                      |                                                               |  |
| $C$ is not included         | ${}^3C_1 \times {}^8C_4 \times {}^3C_2 \times {}^5C_4 = 3150$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                              |                 |                             |                                                                    |                             |                                                                     |                      |                                                               |  |
| $C$ plays defender          | ${}^3C_1 \times {}^8C_3 \times {}^3C_2 \times {}^5C_4 = 2520$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                              |                 |                             |                                                                    |                             |                                                                     |                      |                                                               |  |
| $C$ plays midfielder        | ${}^3C_1 \times {}^8C_4 \times {}^3C_1 \times {}^5C_4 = 3150$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                              |                 |                             |                                                                    |                             |                                                                     |                      |                                                               |  |
| 7i)                         | Let $X$ be the number of sixes obtained in 10 rolls of the fair die. Then $X \sim B\left(10, \frac{1}{6}\right)$<br>$P(X = 3) = 0.155$ (to 3 s.f.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                              |                 |                             |                                                                    |                             |                                                                     |                      |                                                               |  |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                              |
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| 7ii)  | <p>Let <math>Y</math> be the number of sixes obtained in 60 rolls of the fair die. Then <math>Y \sim B\left(60, \frac{1}{6}\right)</math></p> <p>Since <math>n = 60</math> is sufficiently large and <math>np = 10 &gt; 5</math> and <math>n(1-p) = 50 &gt; 5</math>,</p> <p><math>Y \sim N\left(10, 60\left(\frac{1}{6}\right)\left(\frac{5}{6}\right)\right)</math> approximately</p> <p><math>P(5 \leq Y \leq 8) = P(4.5 &lt; Y &lt; 8.5)</math><br/> <math>= 0.273</math> (to 3 s.f.)</p> | Not in H2 Math 9758 syllabus |
| 7iii) | <p>Let <math>W</math> be the number of sixes obtained in 60 rolls of the biased die. Then <math>W \sim B\left(60, \frac{1}{15}\right)</math></p> <p>Since <math>n = 60</math> is sufficiently large and <math>np = 4 &lt; 5</math>,</p> <p><math>W \sim \text{Po}(4)</math> approximately</p> <p><math>P(5 \leq Y \leq 8) = P(Y \leq 8) - P(Y \leq 4)</math><br/> <math>= 0.350</math> (to 3 s.f.)</p>                                                                                        | Not in H2 Math 9758 syllabus |
| 8a    | <p>(i)</p>  <p>(ii)</p>                                                                                                                                                                                                                                                                                                  |                              |
| 8bi)  | <p>The product moment correlation coefficient between <math>m</math> and <math>P</math> is <math>r_1 \approx -0.947045 \approx -0.9470</math> (to 4 d.p.)</p> <p>The product moment correlation coefficient between <math>\ln m</math> and <math>P</math> is <math>r_2 \approx -0.97492 \approx -0.9749</math> (to 4 d.p.)</p>                                                                                                                                                                |                              |
| 8bii) | <p>Since <math> r_2  &gt;  r_1 </math><br/> <math>\therefore P = c \ln m + d</math> is the better model than <math>P = am + b</math>.</p> <p>The equation of the regression line of <math>P</math> on <math>\ln m</math> is<br/> <math>P = 195693.56 - 33659.73 \ln m</math><br/> i.e. <math>P = 196000 - 33700 \ln m</math> (to 3 s.f.)</p>                                                                                                                                                  |                              |

|        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                     |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| 8biii) | When $m = 50$ , $P \approx 64015.92$ . Hence estimated price is \$64000 (nearest 100)                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                     |
| 9)     | Let $T$ be the number of minutes that the bus is late in a day under the control of the new company.<br>Let $\mu$ minutes be the population mean number of minutes that the bus is late.                                                                                                                                                                                                                                                                                                                                                |                                                                                                     |
| 9i)    | $H_0 : \mu = 4.3$<br>$H_1 : \mu < 4.3$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                     |
| 9ii)   | $k^2 = 3.2$<br>$\Rightarrow$ unbiased estimate of population variance,<br>$s^2 = \frac{n}{n-1} k^2 = \frac{64}{19}$ Level of significance: 10%<br><br>Test Statistic:<br>When $H_0$ is true, $Z = \frac{\bar{X} - 4.3}{S/\sqrt{20}}$<br><br>For $H_0$ not to be rejected, $z > -1.28155$<br>$\therefore \frac{\bar{t} - 4.3}{\sqrt{64/380}} > -1.28155 \Rightarrow \bar{t} > 3.7741$<br>Required set is $\{\bar{t} \in \mathbb{R} : \bar{t} > 3.77\}$ | <p>Not in H2 Math 9758 syllabus</p> <p>Try the question again with <math>n = 30</math> instead.</p> |
| 9iii)  | $\bar{t} = 4.0$ , $n = 20$ , $s^2 = \frac{20}{19} k^2$<br>$\therefore z = \frac{4.0 - 4.3}{\sqrt{\frac{20k^2}{19}}/\sqrt{20}}$<br>$H_0$ is rejected $\Rightarrow z \leq -1.28155$<br>$\therefore \frac{4.0 - 4.3}{\sqrt{\frac{20k^2}{19}}/\sqrt{20}} \leq -1.28155$<br>$\Rightarrow \frac{-0.3\sqrt{19}}{\sqrt{k^2}} \leq -1.28155$<br>$\Rightarrow \sqrt{k^2} \leq 1.02038$<br>$\Rightarrow k^2 \leq 1.0412$<br>Required set is $\{k^2 \in \mathbb{R} : 0 < k^2 \leq 1.04\}$                                                           | <p>Not in H2 Math 9758 syllabus</p> <p>Try the question again with <math>n = 20</math> instead</p>  |
| 10ia)  | Required probability<br>$= \frac{1}{10} \times \frac{2}{10} \times \frac{1}{10} = 0.002$                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                     |

| 10ib)             | Required probability<br>$= 1 - \frac{9}{10} \times \frac{8}{10} \times \frac{9}{10}$ $= 0.352$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                              |                          |                   |                                                                |                   |                                                                |                   |                                                                |                              |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|--------------------------|-------------------|----------------------------------------------------------------|-------------------|----------------------------------------------------------------|-------------------|----------------------------------------------------------------|------------------------------|
| 10ic)             | <table border="1"><thead><tr><th>Display</th><th>Probability</th></tr></thead><tbody><tr><td><math>\times \times +</math></td><td><math>\frac{3}{10} \times \frac{1}{10} \times \frac{2}{10} = 0.006</math></td></tr><tr><td><math>\times + \times</math></td><td><math>\frac{3}{10} \times \frac{3}{10} \times \frac{4}{10} = 0.036</math></td></tr><tr><td><math>+ \times \times</math></td><td><math>\frac{4}{10} \times \frac{1}{10} \times \frac{4}{10} = 0.016</math></td></tr></tbody></table><br>Required probability = $0.006 + 0.036 + 0.016$<br>$= 0.058$                                                                                                                                                                                                                                                                                                                 | Display                      | Probability              | $\times \times +$ | $\frac{3}{10} \times \frac{1}{10} \times \frac{2}{10} = 0.006$ | $\times + \times$ | $\frac{3}{10} \times \frac{3}{10} \times \frac{4}{10} = 0.036$ | $+ \times \times$ | $\frac{4}{10} \times \frac{1}{10} \times \frac{4}{10} = 0.016$ |                              |
| Display           | Probability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                              |                          |                   |                                                                |                   |                                                                |                   |                                                                |                              |
| $\times \times +$ | $\frac{3}{10} \times \frac{1}{10} \times \frac{2}{10} = 0.006$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                              |                          |                   |                                                                |                   |                                                                |                   |                                                                |                              |
| $\times + \times$ | $\frac{3}{10} \times \frac{3}{10} \times \frac{4}{10} = 0.036$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                              |                          |                   |                                                                |                   |                                                                |                   |                                                                |                              |
| $+ \times \times$ | $\frac{4}{10} \times \frac{1}{10} \times \frac{4}{10} = 0.016$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                              |                          |                   |                                                                |                   |                                                                |                   |                                                                |                              |
| 10ii)             | $P(\text{the other 2 symbols are + and o} \mid \text{exactly one of the symbols is } *)$ $= \frac{P(\text{the 3 symbols are } * + \text{ o in any order})}{P(\text{exactly one of the symbols is } *)}$ $= \frac{\left(\frac{4}{10} \times \frac{4}{10} \times \frac{1}{10}\right) + \left(\frac{4}{10} \times \frac{2}{10} \times \frac{3}{10}\right) + \left(\frac{2}{10} \times \frac{3}{10} \times \frac{1}{10}\right) + \left(\frac{2}{10} \times \frac{2}{10} \times \frac{2}{10}\right) + \left(\frac{1}{10} \times \frac{3}{10} \times \frac{3}{10}\right) + \left(\frac{1}{10} \times \frac{4}{10} \times \frac{2}{10}\right)}{\left(\frac{1}{10} \times \frac{8}{10} \times \frac{9}{10}\right) + \left(\frac{9}{10} \times \frac{2}{10} \times \frac{9}{10}\right) + \left(\frac{9}{10} \times \frac{8}{10} \times \frac{1}{10}\right)}$ $= \frac{71}{306} \approx 0.232$ |                              |                          |                   |                                                                |                   |                                                                |                   |                                                                |                              |
| 11)               | Let $X$ and $Y$ be the number of original paintings and prints sold per week respectively.<br>Then $X \sim \text{Po}(2)$ and $Y \sim \text{Po}(11)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Not in H2 Math 9758 syllabus |                          |                   |                                                                |                   |                                                                |                   |                                                                |                              |
| 11ia)             | $P(Y > 8) = 1 - P(Y \leq 8)$<br>$= 0.768$ (to 3 s.f.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Not in H2 Math 9758 syllabus |                          |                   |                                                                |                   |                                                                |                   |                                                                |                              |
| 11ib)             | $X + Y \sim \text{Po}(13)$<br>$P(X + Y < 15) = P(X + Y \leq 14)$<br>$= 0.675$ (to 3 s.f.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Not in H2 Math 9758 syllabus |                          |                   |                                                                |                   |                                                                |                   |                                                                |                              |
| 11ii)             | $S = X_1 + X_2 + \dots + X_n \sim \text{Po}(2n)$<br>$P(S < 3) < 0.01 \Rightarrow P(S \leq 2) < 0.01$<br>$\Rightarrow e^{-2n} + (2n)e^{-2n} + \frac{(2n)^2}{2!}e^{-2n} < 0.01$<br>$\Rightarrow e^{-2n}(1 + 2n + 2n^2) < 0.01$ <table border="1"><thead><tr><th><math>n</math></th><th><math>e^{-2n}(1 + 2n + 2n^2)</math></th></tr></thead><tbody><tr><td>4</td><td><math>0.01375 &gt; 0.01</math></td></tr><tr><td>5</td><td><math>0.00277 &lt; 0.01</math></td></tr><tr><td>6</td><td><math>5.2 \times 10^{-4} &lt; 0.01</math></td></tr></tbody></table><br>Smallest possible integer value of $n$ is 5.                                                                                                                                                                                                                                                                           | $n$                          | $e^{-2n}(1 + 2n + 2n^2)$ | 4                 | $0.01375 > 0.01$                                               | 5                 | $0.00277 < 0.01$                                               | 6                 | $5.2 \times 10^{-4} < 0.01$                                    | Not in H2 Math 9758 syllabus |
| $n$               | $e^{-2n}(1 + 2n + 2n^2)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                              |                          |                   |                                                                |                   |                                                                |                   |                                                                |                              |
| 4                 | $0.01375 > 0.01$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                              |                          |                   |                                                                |                   |                                                                |                   |                                                                |                              |
| 5                 | $0.00277 < 0.01$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                              |                          |                   |                                                                |                   |                                                                |                   |                                                                |                              |
| 6                 | $5.2 \times 10^{-4} < 0.01$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                              |                          |                   |                                                                |                   |                                                                |                   |                                                                |                              |

|        |                                                                                                                                                                                                                                                                      |                              |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| 11iii) | $T = Y_1 + Y_2 + \dots + Y_{52} \sim \text{Po}(52 \times 11)$<br>i.e. $T \sim \text{Po}(572)$<br>Since $\lambda = 572 > 10$ ,<br>$T \sim N(572, 572)$ approximately<br>$P(T > 550) = P(T > 550.5)$<br>$= 0.816$ (to 3 s.f.)                                          | Not in H2 Math 9758 syllabus |
| 11iv)  | Since sales will increase during weekends and public holidays, hence average number of sales for the original and prints would not be constant from one day to another day and also the sales would not be independent. Hence the assumptions made may not be valid. | Not in H2 Math 9758 syllabus |